

北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 2 - Student

学院 School	Queen Mary School Hainan	专业 Programme	Information and Computing Sciences (ICS)		
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论文题目 Project Title	Network Analysis of the Chinese High-Speed Railway System				
论文概述 Project outline Write about 500-800 words Minimum 3 References Please refer to Project Student Handbook section 3.2	With the continuous progress of modern technology, China's high-speed railway network—recognized as one of the most extensive and sophisticated transport systems worldwide—has evolved into a fundamental infrastructure supporting regional coordination and improving national mobility efficiency. Its spatial structure demonstrates significant regional diversity: in the economically advanced eastern provinces, dense urban clusters and high travel demand have formed a tightly connected, multi-path network; whereas in the western regions, where terrain is complex and population density is low, railway accessibility remains relatively sparse. From a data science perspective, this study conceptualizes the Chinese high-speed railway as a complex spatial network, where stations act as nodes and track segments serve as edges. Using systematic network analysis, the research aims to reveal the system's structural patterns, operational behaviors, and potential optimization strategies. These insights will offer both theoretical value and practical guidance for scientific railway planning and efficient system management. The central goal of the project is to establish a comprehensive and accurate model of the national high-speed rail system—scalable to specific provinces or regional networks when needed. To ensure the model's reliability, several dimensions must be met. • Accuracy: precisely represent the spatial distribution of nodes, direction of routes, and their real-world connections, while minimizing mapping errors. • Structural logic: maintain consistency with topological principles of spatial networks, clearly defining the attributes and interrelations of nodes and edges. • Internal consistency: unify data standards, computational criteria, and modeling logic to prevent contradictions. • Applicability: adjust the scope based on research focus, ensuring that crucial nodes and links are included without introducing redundant data. The data used in this study come from two primary sources. First, part of the dataset is derived from the work of Lin Ma and Yang Yang, who examined the distributional effects of China's transportation networks. When using these data, the original study is cited as: Lin, M., & Yang, Y. (2024). Distributional impacts of China's transportation networks. <i>Journal of International Economics</i>, 103873. Second, additional urban transportation information is obtained from the ChinaCitiesV1 platform (http://chinacitiesv1.parisgeo.cnrs.fr/)				

	), maintained by CNRS Paris Geography, which provides openly accessible and well-documented spatial datasets on Chinese cities. This model forms the foundation for all subsequent analyses, serving as a dependable framework for in-depth exploration of the railway network's structural and dynamic properties. Methodologically, the research develops a mathematical framework to address the project's key questions. It focuses on three primary areas: 1. Temporal evolution of network characteristics – by comparing data from different years, variations in indicators such as node density, connectivity strength, mean path length, and clustering coefficient will be examined to capture the network's development trajectory. 2. The study will conduct a detailed analysis of the topological characteristics of the Chinese high-speed railway network. The network will be evaluated using a range of classical and spatial network metrics to reveal its structural organization and functional properties. In this framework, the average degree will reflect the overall connectivity level of the system. the analysis will incorporate betweenness and closeness centrality to describe the relative importance of stations within the network's flow structure. 3. Robustness and resilience – network robustness will be evaluated through connectivity changes under simulated node failures. Random failures will be generated using probabilistic deletion, while intentional disruptions will target nodes by importance. Network resilience, representing the retained connectivity after failure, will be quantified using the <i>resilience triangle</i> framework proposed by Bruneau et al.
	References [1] Jiao, J. J., Wang, J. E., & Jin, F. J. (2017). <i>Impacts of high-speed rail lines on the city network in China</i>. Journal of Transport Geography, 60, 257–266. https://doi.org/10.1016/j.jtrangeo.2017.03.010 [2] Zhang, D. M., Du, F., Huang, H. W., Zhang, F., Ayyub, B. M., & Beer, M. (2018). <i>Resiliency assessment of urban rail transit networks: Shanghai metro as an example</i>. Safety Science, 106, 230–243. https://doi.org/10.1016/j.ssci.2018.03.023 [3] Chen, Z. X., Chen, J. H., Zheng, H., & Gao, T. Z. (2025). <i>Enhancing high-speed railway timetable resilience: A two-level spatiotemporal network model focused on disturbance absorption</i>. Urban Rail Transit, 11, 67–89. https://doi.org/10.1007/s40864-024-00235-3 [4] Bruneau, M., Chang, S. E., Eguchi, R. T., Lee, G. C., O'Rourke, T. D., Reinhorn, A. M., Shinozuka, M., Tierney, K., Wallace, W. A., & von Winterfeldt, D. (2003). <i>A framework to quantitatively assess and enhance the seismic resilience of communities</i>. Earthquake Spectra, 19, 733–752.
道德规范 Ethics	Please confirm by checking the box: ☒ I confirm that I have discussed ethical issues with my supervisor.

Please discuss ethical issues with your supervisor. Please refer to Project Student Handbook section 4.1	Summary of ethical issues: All data employed in this research are publicly available secondary data, involve no individual or sensitive personal information, and therefore raise no ethical concerns regarding privacy, human subjects, or field experimentation.
中期目标 Mid-Term Target. It must be a measurable/tangible outcome, E.g. prototype of software, hardware or simulation. This will be the criteria for Mid-Term Oral Exam (Viva).	The mid-term objective of this project is to establish a complete and well-structured model of China's high-speed railway network that accurately reflects the spatial distribution of stations and the connections between them. Based on this network model, a comprehensive topological analysis will be conducted, including the examination of degree characteristics, centrality measures, shortest-path structures, and overall connectivity patterns. A preliminary assessment of the system's robustness will also be carried out as part of the mid-term objectives. Achieving this stage goal will provide a solid analytical foundation for subsequent studies on network evolution, robustness, and system resilience.

Work Plan (Gantt Chart)

Fill in the sub-tasks and insert a letter X in the cells to show the extent of each task

	Nov 1-15	Nov 16-30	Dec 1-15	Dec 16-31	Jan 1-15	Jan 16-31	Feb 1-15	Feb 16-28	Mar 1-15	Mar 16-31	Apr 1-15	Apr 16-30
Task 1 Review relevant literature and software on transport networks												
Read 10 core papers on railway transport (including specific ones on China)	X	X	X									
Identify relevant Python packages and GitHub repositories for transport network analysis		X	X									
Summarize key findings from the literature and their relevance for the project			X	X								
Task 2 Collect and curate datasets on the population of major Chinese cities and high-speed railways.												
Review potential datasets and their use in the literature	X	X	X									
Assess completeness and bias of datasets for suitability	X	X	X	X								
Describe the datasets alongside their limitations in the report				X	X							
Task 3 Design a methodology inspired by existing transport models												
Review existing methods from the literature					X	X						
Combine methods to create a methodology based on the research question					X	X	X	X				
Describe the methodology and its derivation in the report							X	X				
Task 4 Develop a rigorous, legible, and reproducible computational implementation of the methodology.												
Implement the methodology and run experiments						X	X	X	X			
Describe the implementation in the report									X	X		
Report the results, alongside visuals, in the report									X	X		

Deposit the cleaned, well-commented, reproducible code on GitHub											X	X
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